



IPE 332: Production Processes Sessional

Experiment 1: Study of Chips and Cutting Zone Temperature in Turning Medium Carbon Steel by Uncoated Carbide Insert

Handout

Prepared by,

Mostofa Habib Fardin

Lecturer

Dept. of IPE, BUET

1. Motivation

Three practical concerns:

1. **Chip form affects safety and productivity on the shop floor:** A long, uncontrolled ribbon or spiral chip can wrap around the tool, the workpiece, or the operator's hand. In production, chip form is controlled deliberately (through chip breakers, feed selection, tool geometry) so that chips break into short, manageable segments.
2. **Chip geometry is a direct, measurable signature of the deformation happening in the cut:** The layer of metal is thicker after it leaves the tool than it was before the cut. This thickening is not incidental; it is a consequence of plastic shear deformation ahead of the tool edge. The ratio of these two thicknesses, the chip reduction coefficient.
3. **Cutting temperature governs tool life and part quality:** Almost all of the mechanical work done in cutting converts to heat, concentrated in a very small volume at the tool tip. This heat softens the tool material, accelerates chemical wear mechanisms (you will study these in Experiment 2), and can distort the workpiece dimensionally if not controlled.

2. Theory

2.1 Mechanics of Chip Formation

When the cutting tool is fed into the rotating workpiece, the layer of material directly ahead of the tool edge does not simply detach. It first deforms elastically, then, as stress builds beyond the material's yield point, deforms plastically. Plastic deformation is concentrated in a narrow zone extending from the tool's cutting edge. This is called shear zone.

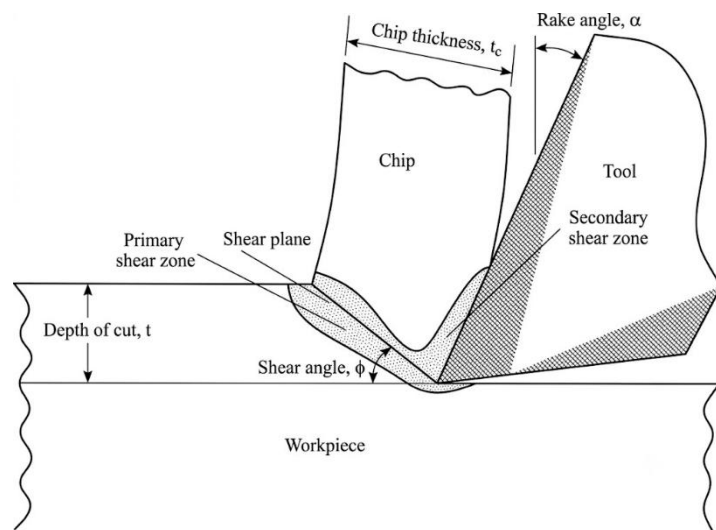


Figure 1: Chip Formation

A second, thinner zone of deformation exists at the interface between the underside of the chip and the tool's rake face, called the **secondary deformation zone**. Here, friction between the moving chip and the stationary tool face causes additional shearing, further heating the chip

and the tool. Heat is generated at the shear plane (from bulk plastic deformation) and at the tool-chip interface (from friction), and the temperature you measure at the tool tip in this experiment is the net result of both sources.

Once past the shear plane, the deformed material is now the **chip**. Because it has thickened relative to its original (undeformed) thickness and because friction along the tool face resists its flow, the chip does not slide off freely, so it curls and depending on the material's ductility and the cutting conditions, it either flows continuously or fractures repeatedly into segments. This behavior is the basis for the classification of chip types taken up in the next section.

2.2 Classification of Chip Types

Depending on the workpiece material, cutting conditions, and tool geometry, the chip produced during machining falls into one of four broad categories. These are distinguished by how the material fails as it crosses the shear plane, and by the resulting chip's continuity, surface texture, and mechanical behavior.

1. **Discontinuous Chips:** They form as small, loosely-connected segments. They occur when the workpiece material has limited ductility and fractures repeatedly ahead of the tool rather than flowing plastically along the shear plane. Hard, brittle materials such as cast iron, bronze, and brass typically produce this chip type, even under otherwise favorable cutting conditions. Because each segment separates cleanly, discontinuous chips are easy to clear from the cutting zone, but the intermittent fracturing causes fluctuating cutting forces and generally leaves a rougher machined surface.
2. **Continuous Chips:** They form as long, unbroken ribbons or coils. The material deforms plastically and flows smoothly up the rake face without fracturing, which is characteristic of ductile materials cut under stable conditions (moderate to high speed, sharp tool, effective lubrication). The underside of the chip (the side that was in contact with the tool) is smooth, while the outer side shows fine striations from the intermittent slip within the shear zone. This mode gives steady cutting forces and the best surface finish of the four types, but poses a practical hazard: an unbroken chip can coil into a tangled mass around the tool and workpiece which is known in industry as a "**bird's nest**". It can rub against and damage the machined surface, block coolant from reaching the cutting zone, and even get re-cut by the tool.
3. **Continuous chip with built-up edge (BUE):** It occurs when small particles of the workpiece material weld themselves to the tool's rake face near the cutting edge under high

pressure and temperature, particularly at low-to-moderate cutting speeds. This built-up material periodically grows, breaks off, and re-forms. Some of the broken fragments adhere to the chip underside and some to the newly machined surface, producing a chip that is continuous but visibly less smooth than a true continuous chip. The effect on the tool is genuinely two-sided: research shows that at low cutting speeds a stable BUE can shield the rake face and cutting edge, measurably **reducing flank and crater wear** in that speed range. But the same fragments are hard and abrasive, and when they break away, they **degrade the machined surface finish and can drag across the flank face, causing irregular wear**. In practice, BUE is treated as an undesirable condition to be avoided (usually by raising cutting speed or improving lubrication) rather than a wear-prevention strategy to be relied upon.

4. **Serrated Chips:** They show a saw-tooth appearance caused by cyclic variation between a zone of high shear strain and a zone of low shear strain within the chip, effectively a semi-continuous chip. This is most closely associated with difficult-to-machine materials such as titanium alloys, nickel-base superalloys, and austenitic stainless steels, but can also appear in ordinary steels when cut at high speed. The mechanism is essentially localized, **cyclic tool chatter driven by force variation between the high-strain and low-strain phases**, sometimes described as micro-chipping of the material at the shear zone.

2.2.1 Which chip type is preferred?

No chip type is better without a trade-off. **Continuous chip formation is mechanically the smoothest**; steady shear, steady cutting force, and the best surface finish of the four types. But an uncontrolled continuous chip is a practical liability: it tends to coil up around the tool and workpiece ("bird's nesting"), which blocks coolant from reaching the cutting zone and can even cause the tool to re-cut its own chip, adding heat rather than removing it. **Discontinuous chips, by contrast, break and clear away from the cutting zone immediately, allowing coolant to reach the tool-chip interface directly** and giving better temperature control at the point of cutting, even though they arise from a less ductile or less stable cutting condition.

This is precisely why chip breakers are standard on production tooling: they force an otherwise continuous chip to curl tightly and fracture into short segments, combining the smooth shearing mechanics of continuous chip formation with the cooling and evacuation advantage of a short, discontinuous-like chip. In short, the practically preferred outcome is **a continuous chip broken into short segments**

2.2.2 Summary Table: Chip Types in Metal Cutting

Chip Type	Formation Mechanism	Typical Conditions / Materials	Surface Finish	Cooling / Evacuation	Overall Assessment
Discontinuous	Repeated brittle fracture ahead of the tool	Brittle materials (cast iron, bronze, brass); ductile materials at very low speed / heavy feed	Poor–Fair	Good/ clears the zone easily	Easy to handle, but signals brittleness or unstable cutting
Continuous	Smooth plastic flow along the shear plane, no fracture	Ductile materials (steel, aluminum) at moderate–high speed, sharp tool	Good	Poor if uncontrolled/ bird's-nests, blocks coolant	Best mechanics; needs a chip breaker to be usable
Continuous with BUE	Workpiece material welds to and periodically breaks from the rake face	Ductile materials at low–moderate speed	Poor (fragmented, rough)	Fair	Undesirable overall: reduces flank/crater wear at low speed in some conditions, but degrades finish; avoided in practice
Serrated / segmented	Adiabatic shear localization (thermal softening outpaces strain hardening)	Titanium alloys, nickel-base superalloys, austenitic stainless steel; also ordinary steel at high speed	Fair–Poor	Fair/ Poor	Undesirable — signals a difficult-to-machine condition

2.3 Chip Shapes in Practice

Chip shape is governed primarily by feed rate, cutting speed, depth of cut, and tool/chip-breaker geometry, and secondarily by workpiece ductility and tool-chip friction along with other factors. The same workpiece material can produce very different chip shapes purely by changing feed and speed, which is what you should expect to observe across the parameter combinations in this experiment.

2.3.1 Which chip shape is preferred?

Lab manual gives an explicit ranking for this experiment: **spiral coil chips are the most desired in finish machining**, because they break off on their own once they reach a certain length, keeping the cut stable; **cracking chips are the least desired**, and **C-shape chips are next-to-least desired**, because most C-shape chips separate only by colliding with the tool's flank face or the workpiece surface, which disturbs cutting stability and surface finish.

It is worth flagging one nuance here, since you may see a different ranking elsewhere. In modern CNC turning practice, tightly-curved "6" or "9" shaped chips, essentially short C-shapes are often described as the *ideal* outcome, provided they snap under their own momentum against a non-critical surface (the part shoulder, the chip pan) rather than against the cutting edge or the finished surface, and provided they are not curled so tightly that they trap heat and pressure (a real problem in tough alloys). In other words, the disagreement is not really about the shape itself, but about *how and where* it breaks: a C-shape chip that fractures cleanly away from critical surfaces is fine; one that only breaks by repeatedly striking the tool or the workpiece is not. For this experiment, follow your manual's ranking (spiral best, C-shape and cracking chips least desired) since that reflects the specific tool/material combination and finishing context being tested here.

2.4 Chip Color as a Temperature and Environment Indicator

Chip color is a direct visual record of the temperature the chip reached during cutting. The mechanism is [thin-film interference](#): as a freshly-cut, bare steel surface is heated, it oxidizes, forming a thin iron-oxide layer. This layer is thin enough that light passing through it reflects off both its outer surface and the underlying metal, and the two reflected waves interfere. As temperature rises, the oxide layer grows thicker, which shifts the interference pattern and therefore the visible color. The sequence as temperature increases: **metallic** → **pale**

straw/yellow → **brown** → **purple** → **blue** → **burnt blue** → **black**, with color deepening further toward near-black at the highest temperatures.

A shop-floor rule of thumb captures the practical use of this: **blue or purple chips are generally a good sign** (they show that the heat generated in cutting is being carried away in the chip rather than staying in the tool or the workpiece), while pale, barely-colored chips at high material-removal rates can indicate the tool isn't shedding heat effectively, and very dark, near-black chips indicate excessive heat.

2.4.1 Why do wet and dry conditions produce different chip colors?

In dry cutting, almost all the heat generated at the shear plane and the tool-chip interface is absorbed by the chip, the tool, and the workpiece, with no external heat removal. So, the chip's final oxide color reflects the full temperature it reached. Introducing a cutting fluid (as in the wet-condition) actively removes heat from the cutting zone during chip formation, lowering the temperature the chip reaches before it separates. Additionally, it shields the chip from oxygen. This is why, at matched speed and feed, wet-condition chips typically show lighter colors (silver, golden, light straw) than dry-condition chips (blue, purple or burnt/dark tones)

2.5 Chip Reduction Coefficient (ξ)

2.5.1 Definition

The chip that leaves the cut is thicker than the layer of material it came from. This is because the material deforms (compress) due to the force exerted on it in front of the tool. If a_1 is the uncut chip thickness (the thickness of the layer before it is removed) and a_2 is the chip thickness (after deformation), the **chip reduction coefficient** is defined as:

$$\xi = \frac{a_2}{a_1}$$

Since the chip always thickens during cutting, $a_2 > a_1$ **always**, so ξ is **always greater than 1**. Its reciprocal, $r = a_1/a_2 = 1/\xi$, is called the **cutting ratio**. Cutting ratio is always less than 1. Both quantities describe the same physical phenomenon; ξ is simply the more commonly used form in this context because it scales in the same direction as cutting force and deformation severity; a larger ξ means more deformation, not less.

In this experiment, a_1 is not measured directly but computed from the feed rate and the tool's principal cutting edge angle φ :

$$a_1 = S_0 \sin \phi$$

so that:

$$\xi = \frac{a_2}{a_1} = \frac{a_2}{S_0 \sin \phi}$$

where a_2 is measured directly (with slide calipers, on the collected chip), S_0 is the feed rate (mm/rev), and ϕ is a fixed property of the tool geometry used.

2.5.2 Physical meaning

ξ is a direct, measurable index of how much plastic deformation the material underwent while forming the chip. A higher ξ means the chip thickened more relative to the layer it came from, which corresponds to a smaller shear angle ϕ . A larger shear plane area requires a larger shear force to separate the chip, and that force translates directly into higher cutting force, higher power consumption, and more heat generated in the cut. **In short: a lower ξ is generally the better outcome;** it means the material sheared more efficiently, with less force and less energy wasted as heat.

This connects ξ to the shear-angle relation covered in your theory classes (the same one behind Merchant's equation): as the shear angle ϕ increases, the chip thickness ratio increases and ξ decreases. Anything that increases the shear angle such as, a larger (more positive) rake angle, lower tool-chip friction, tends to lower ξ . Anything that reduces the shear angle such as, a more negative or less positive rake angle, higher friction, a larger tool nose radius, tends to raise it.

2.5.3 Effect of process parameters

Based on both the theory above and results reported for this exact experiment, you should expect:

- **ξ decreases as feed rate increases.** At a fixed shear angle, $a_1 = S_0 \sin \phi$ grows directly with feed, and this growth in the denominator outpaces the corresponding growth in a_2 , so the ratio $\xi = a_2/a_1$ falls. This is the most consistent and strongest trend across the reference data reviewed for this experiment.
- **ξ is comparatively less sensitive to cutting speed, though a mild decreasing trend is often observed,** particularly at lower feed rates which can be attributed to reduced tool-chip friction and a corresponding rise in shear angle as speed increases.

- ξ is generally lower under wet cutting than dry cutting at matched speed/feed, since the cutting fluid reduces friction at the tool-chip interface, raising the shear angle.

The *direction* of these trends (ξ falls with feed, mildly falls with speed, falls with cooling) is consistent and robust across the literature and prior batch data for this experiment.

2.6 Cutting-Zone Temperature and the Tool-Work Thermocouple Method

Why it matters: most of the mechanical work of cutting converts to heat at the shear plane and tool-chip interface. This heat governs tool wear rate, dimensional accuracy, and chip color.

Principle: the tool and workpiece, being dissimilar conductors in contact at the cutting zone, form a natural thermocouple junction. The EMF generated is picked up via a copper brush (maintaining contact with the rotating workpiece) and read on a millivoltmeter. Insulation isolates the circuit from the machine frame so the reading reflects only the tool-work junction.

Calibration, not a universal law: the relation used in this experiment,

$$\theta = 75.28 + 63.05 \text{ mV} - 0.57 \text{ mV}^2$$

is an empirical curve fitted specifically for this tool-work material pair. Hence, it cannot be reused for a different tool/workpiece combination without recalibration.

Limitation to note: The readings can be affected by contact resistance and circuit isolation quality, a source of experimental error if insulation is imperfect.